

Pedestrian Trajectory Prediction and World Modelling for Safe Robot Navigation in Human Crowds

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Abstract—Safe robot navigation in human crowds depends on predicting uncertain pedestrian motion before planning or runtime safety filtering can make reliable decisions. Building on our earlier survey, which identified a gap between open-loop prediction quality and closed-loop navigation safety, this report presents a prediction-centered stack with three connected parts: synthetic trajectory augmentation, a residual crowd world model, and a benchmark of trajectory predictors. The implementation includes a Social-CVAE generator pretrained on SDD and fine-tuned on ETH/UCY, a lightweight world model that predicts pedestrian futures as a learned residual over constant velocity, and five predictors: Social-LSTM, SLSTM+V, Social-GRU, a Trajectory Transformer, and a diffusion predictor, evaluated against constant velocity and Trajectron++. On held-out eth, the Simple Residual Model improves ADE from 1.0755 m to 0.9514 m and FDE from 2.2819 m to 1.9171 m. On ETH/UCY, the Transformer gives the lowest ADE, 0.568 m, and NLL, 1.54, while Trajectron++ gives the lowest minADE@20, 0.385 m. SDD pretraining improves recurrent models by 5 to 10 percent. Filtered synthetic data supports augmentation but requires quality control.

Index Terms—trajectory prediction, world models, pedestrian motion, synthetic data, transformers, diffusion models, crowd navigation

I. INTRODUCTION

Mobile robots operating in corridors, hospitals, stations, campuses, or shopping spaces must move through people who are not passive obstacles. People slow down, turn, give way, ignore the robot, or react late. The navigation problem is therefore more than geometric collision avoidance; it requires forecasting uncertain human motion quickly enough for a downstream planner and safety layer to use it.

Our earlier survey identified a recurring tradeoff between three failure modes. A predictor strong on ADE/FDE can still miss the safety-critical future, since those metrics average over typical cases. A predictor with wider sample coverage produces uncertainty sets the planner cannot reduce, and the robot stops. A predictor expressive enough to handle both can also miss the 50–100 ms control budget. This report builds the prediction and world modelling components that sit upstream of the safety layer rather than revisiting the survey.

Vaibhav developed the Social-CVAE synthetic trajectory generator and the residual crowd world model, which corrects

a constant-velocity forecast with a learned residual for planning use. Muhammad implemented and evaluated the predictor benchmark: Social-LSTM, SLSTM+V, Social-GRU, a Trajectory Transformer, and a diffusion predictor, with Trajectron++ as the published baseline, all under SDD pretraining and ETH/UCY fine-tuning. Benjamin developed the downstream conformal-and-PPO safety layer. Closed-loop results for the safety layer are not in this repository, so it is described as the integration target rather than an evaluated component.

The report contributes three results. Filtered Social-CVAE samples augment ETH/UCY without degrading downstream prediction accuracy. A residual-on-constant-velocity world model lowers ADE by 11.5% and FDE by 16.0% over the constant-velocity baseline on held-out eth. Five trajectory predictors are compared under a shared protocol on ETH/UCY and SDD; under NLL and best-of-20 coverage the ranking differs sharply from the ADE ranking, and SDD pretraining reduces recurrent ADE by 5 to 10% while leaving the Transformer unchanged.

A. Design Rationale From The Initial Survey

Three design choices in this report follow directly from the survey. Pedestrian motion is multi-modal: a single observation history is consistent with continuing, slowing, turning, or yielding. A single-mode predictor reports the mean of these futures and discards the alternatives, which is unsafe when the alternatives matter. The benchmark therefore includes both point predictors and probabilistic models (Trajectron++ and a diffusion predictor) and reports best-of-20 metrics alongside ADE and FDE.

Predictor uncertainty must also be usable by the safety layer. A split-conformal shield calibrates on residuals: a sharp mean with a well-shaped σ produces a small safety radius, while a wide or poorly calibrated σ inflates the radius until the planner has no feasible motion. NLL is reported alongside ADE because the downstream layer consumes the predicted distribution, not only the mean.

The third constraint is real time. The predictor runs every control cycle, so any accuracy gain that reduces the achieved control frequency degrades closed-loop behaviour. Constant

TABLE I
PLANNED MILESTONES AND SUBMISSION STATUS.

Weeks	Milestone	Status
1–3	Environment setup, baseline datasets, initial Social-LSTM and constant-velocity tests.	Complete
4–6	Prediction models, including Trajectron++ comparison and model-specific metrics.	Complete
7–8	Synthetic data generation and residual crowd world model experiments.	Complete
9–10	Safety/planning interface and conformal/PPO integration design.	Complete
11	Final comparisons, plots, ablations, and report integration.	Complete
12	Final report, presentation material, and submission cleanup.	Complete

velocity and the residual world model appear in the benchmark for this reason, alongside the larger neural predictors. The comparison evaluates the tradeoff each model makes between point accuracy, calibration, sample diversity, and per-cycle compute, rather than ranking architectures in isolation.

II. PROJECT MANAGEMENT

The project plan defines four deliverables: a prediction module, an interaction-aware planner, a confidence-calibrated safety layer, and an end-to-end evaluation. This report covers the prediction module together with the synthetic data pipeline and residual world model that feed it. The planner and safety layer are designed and partially implemented; closed-loop evaluation is scheduled for the next phase and is therefore reported here as an integration target rather than as a measured result.

Vaibhav developed the Social-CVAE generator and the residual world model end-to-end, including the data pipeline, the angle-based filter, and the held-out eth evaluation. Muhammad implemented the predictor benchmark and the transfer-learning protocol: Social-LSTM, SLSTM+V, Social-GRU, the Trajectory Transformer, the diffusion predictor, and Trajectron++ as a baseline, together with SDD pretraining and joint ETH/UCY-plus-synthetic fine-tuning. Benjamin developed the safety layer. It consumes predictor outputs (Gaussian, sample set, or mixture), produces conformal uncertainty regions, and accepts or rewrites the actions proposed by a PPO policy. Closed-loop results for the safety layer are not yet available; the interface nevertheless keeps the prediction work tied to the original safe-navigation objective rather than reducing it to an open-loop benchmark.

Four risks shaped the implementation. Dependency setup was the first: conflicts between the Trajectron++ codebase, the CVAE, and our own training loops were contained by isolating each model in its own loader and not sharing checkpoints across branches. Compute was the second: a single RTX 3090 Ti shared between three students forced leave-one-scene-out splits, early stopping, and an ordering policy that ran smaller variants before larger ones. Integration is the third and is explicitly open, since closed-loop evaluation requires connecting the predictors to Benjamin’s safety layer. The fourth is methodological: heavier models do not always outperform simpler baselines, and the report includes the cases where this

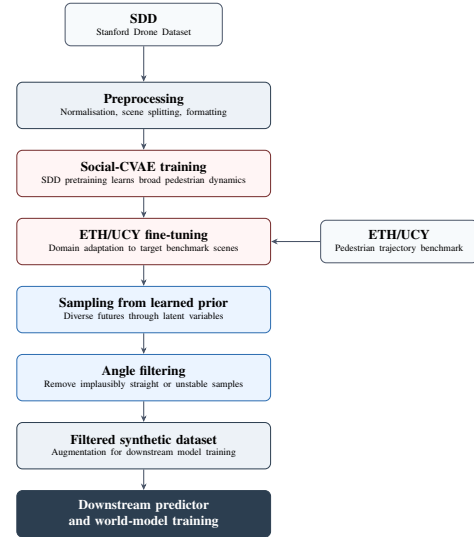


Fig. 1. Synthetic data and world-model pipeline used to support downstream predictor training and eventual safety-aware navigation.

occurred (the Latent Crowd World Model and the diffusion sample-collapse behaviour) rather than omitting them.

A. Deliverable Status

The prediction deliverable is complete: every predictor in the benchmark runs end-to-end with SDD pretraining and joint ETH/UCY-plus-synthetic fine-tuning, and Vaibhav’s filtered synthetic stream feeds the fine-tune stage directly. The residual world model is implemented and evaluated on the held-out eth split; the remaining four ETH/UCY splits are pending and flagged in the limitations. The safety layer is complete at the interface level: predictor residuals and sample sets feed the conformal stage, the PPO policy proposes actions, and the shield can veto. Closed-loop evaluation is the next phase and will report collision rate, time-to-goal, and intervention frequency once the components are connected in simulation.

III. METHODOLOGY AND SYSTEM DESIGN

A. System Overview

Fig. 1 shows the stack used in this report. The first stage expands the data distribution through synthetic trajectory generation. The second stage learns a residual crowd world model for planning-facing pedestrian motion forecasts. The third stage benchmarks several trajectory predictors, some optimized for point accuracy and others for stochastic coverage. These outputs are intended to feed a downstream conformal safety layer, where calibrated prediction errors determine how much space must be reserved around future human positions [1].

B. Synthetic Data Generation

ETH/UCY is the target benchmark, but it contains only five scenes and a limited range of crowd configurations. To increase data scale and motion diversity, we developed a Social-CVAE generator. Synthetic data is used as augmentation, not as a replacement for real trajectories.

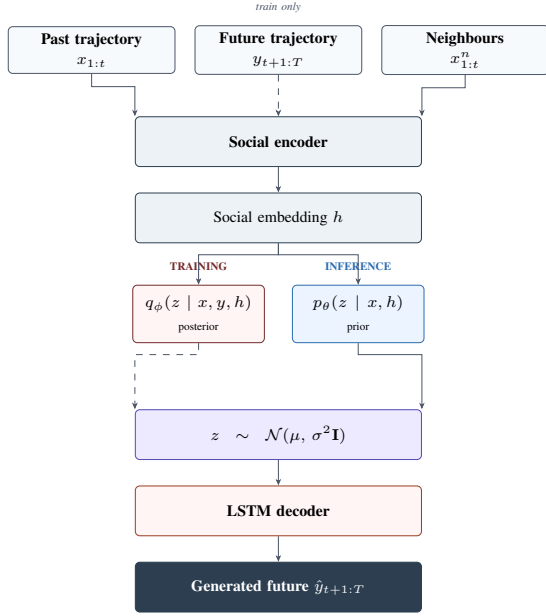


Fig. 2. Social-CVAE generator. The posterior uses future trajectories during training; at generation time, the conditional prior samples futures from the observed trajectory and social context.

The generator follows the standard 8-step observation and 12-step prediction format. It is pretrained on SDD [2], which provides a larger and broader aerial pedestrian dataset, then fine-tuned on ETH/UCY so the generated samples better match the target benchmark distribution. The CVAE structure is appropriate because the same observed path can plausibly lead to several futures: continuing straight, slowing down, turning, or reacting to nearby pedestrians. The latent variable models this ambiguity.

The preprocessing stage converts both SDD and ETH/UCY into fixed-length trajectory windows and normalises them into a shared coordinate convention. Each training sample contains the target pedestrian history, its future trajectory, and nearby pedestrian histories when available. During training, the encoder can condition on the observed future through the approximate posterior $q_\phi(z | x, y, h)$; during generation, only the observed past and social context are available, so samples are drawn from the conditional prior $p_\theta(z | x, h)$. This distinction matters because it prevents the generator from depending on information that would not exist at deployment.

The training objective was

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda_v \mathcal{L}_{\text{vel}} + \beta \mathcal{D}_{\text{KL}}(q_\phi(z) \| p_\theta(z)) + \lambda_s \mathcal{L}_{\text{smooth}}. \quad (1)$$

This combines position reconstruction, velocity consistency, latent regularisation, and trajectory smoothness. For each seed trajectory, we generated 20 candidate futures. Angle-based filtering removes unrealistic samples such as overly straight, overly smooth, or unstable trajectories. This step is important because poor synthetic samples can widen the training distribution away from realistic pedestrian motion.

The filtering stage is part of the model design. Synthetic data should increase support around plausible pedestrian behaviours without making unrealistic curves appear likely to the downstream predictor. Vaibhav’s pipeline keeps synthetic trajectories as a train-only augmentation source: the validation and test sets remain real ETH/UCY scenes, so performance cannot be inflated by testing on generated data.

C. Residual Crowd World Model

The residual crowd world model is a separate contribution from the predictor benchmark. Its purpose is to provide a lightweight planning-facing dynamics model for nearby pedestrian motion. It predicts 12 future timesteps from 8 observed timesteps, matching the ETH/UCY setup, but it does not learn the full trajectory from scratch. Instead, it corrects a physical baseline:

$$\hat{Y} = Y_{\text{CV}} + R_\theta(X, S), \quad (2)$$

where Y_{CV} is the constant-velocity forecast, R_θ is the learned residual, X is the observed target trajectory, and S is optional nearby pedestrian context.

This design choice is conservative. Pedestrians often move smoothly over a few seconds, so constant velocity is a strong prior. The neural network only needs to learn deviations: turning, slowing down, acceleration, or local interaction effects. Trajectories are shifted into a local frame using the last observed position as origin. A GRU encodes the target trajectory; neighbours are encoded and mean-pooled with a validity mask; and the fused representation is decoded into a residual correction.

Three variants separate the benefit of residual learning from the benefit of extra context or capacity. The Simple Residual Model uses only the target pedestrian history and tests whether a learned correction to constant velocity is sufficient. The Social Residual Model adds nearby pedestrian context, testing whether interaction cues improve prediction. The Latent Crowd World Model adds a higher-capacity latent dynamics representation, testing whether a more expressive world model improves generalisation or overfits the small ETH/UCY training split.

The model was trained with

$$\mathcal{L} = \mathcal{L}_{\text{ADE}} + \lambda_f \mathcal{L}_{\text{FDE}} + \lambda_r \mathcal{L}_{\text{res}}. \quad (3)$$

ADE encourages full-horizon accuracy, FDE emphasizes the final position, and residual regularisation discourages unrealistic corrections.

This world model is separate from the larger predictor benchmark. The benchmark asks which forecasting architecture performs best under standard open-loop metrics. The world model asks a planning-facing question: can a lightweight, interpretable dynamics model improve over a hand-designed physical prior while remaining simple enough to use inside a future control loop?

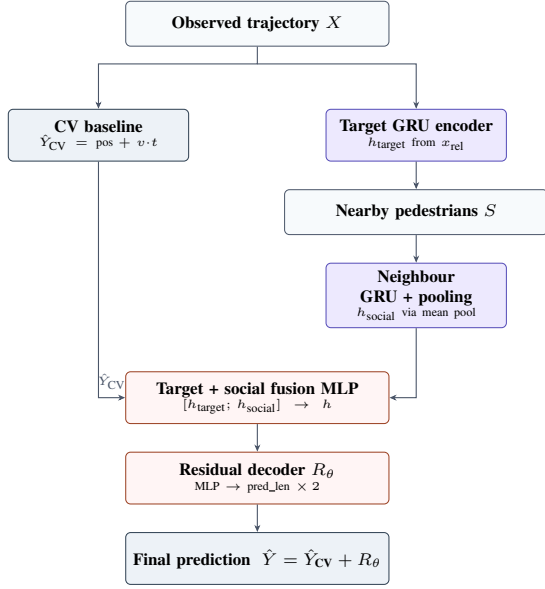


Fig. 3. Residual crowd world model. The final pedestrian forecast combines a constant-velocity prior with a learned residual correction from target and social context.

D. Trajectory Prediction Models

The predictor benchmark evaluates models that output future pedestrian trajectories for downstream planning and safety filtering. Each predictor observes 8 steps at 2.5 Hz (3.2 s) and predicts 12 future steps (4.8 s). Coordinates are represented relative to the last observed position, which reduces dependence on absolute scene location [3]. Social context comes from up to five neighbours inside a 2.0 m radius, with validity masks used when fewer neighbours are present. Except for the constant-velocity and Trajectron++ baselines, models output a bivariate Gaussian $(\mu_x, \mu_y, \sigma_x, \sigma_y, \rho)$ per future step and are trained with negative log-likelihood plus ADE/FDE monitoring. This common interface makes the comparison fair: each model sees the same history and social context, but uses a different inductive bias.

Social-LSTM [3] is the recurrent baseline. It uses an LSTM encoder-decoder with hidden size 128, embedding size 64, and social pooling over nearby pedestrian hidden states. SLSTM+V keeps the same architecture but appends instantaneous velocity

$$v_t = p_t - p_{t-1} \quad (4)$$

to the focal pedestrian input, changing (x_t, y_t) into (x_t, y_t, v_x, v_y) while leaving neighbour inputs as positions. This tests a small but important design question: whether giving the model explicit short-term motion state improves forecasting more reliably than asking the LSTM to infer velocity from positions alone. Social-GRU replaces the LSTM with a bidirectional GRU [4], [5]. Forward and backward states are concatenated and projected into the shared hidden representation, allowing late observation cues, such as a pedestrian

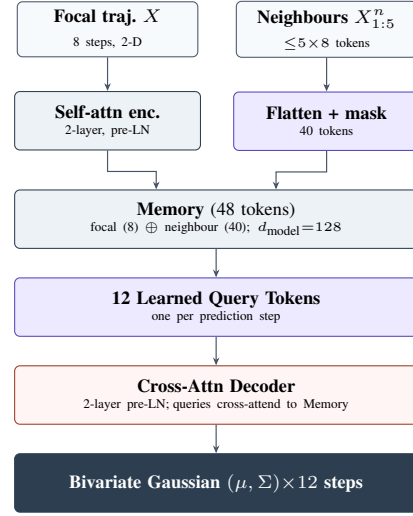


Fig. 4. Trajectory Transformer architecture used in the prediction benchmark.

beginning to decelerate near the end of the history, to influence the encoded trajectory.

The Trajectory Transformer uses a Pre-LN encoder-decoder architecture [6], [7]. Eight focal tokens and up to forty neighbour tokens form the encoder memory; twelve learned decoder queries cross-attend to that memory and output bivariate Gaussian parameters for each future step. This structure gives each predicted timestep direct access to the complete observed social scene.

The diffusion predictor uses a shared Transformer encoder, a Gaussian MLP head, and a denoiser trained with a DDPM loss [8]. A CLS summary feeds the Gaussian head, while the denoiser cross-attends to the encoded trajectory memory and predicts the noise term for a candidate future. The implementation uses 100 diffusion steps, linear $\beta \in [10^{-4}, 0.02]$, DDIM-50 sampling at inference [9], and $K = 20$ samples for diversity metrics. An early implementation detached the Gaussian head from the shared encoder; the final version removes this detach so that the likelihood and denoising losses jointly shape the representation. Trajectron++ [10] is included as the main probabilistic multi-modal baseline because its graph-structured CVAE directly models multi-agent trajectory distributions.

E. Transfer Learning

ETH/UCY is small, so the project uses a two-stage transfer protocol. Stage one pretrains each architecture on SDD for up to 100 epochs with Adam at learning rate 10^{-3} . Stage two fine-tunes each checkpoint on the four ETH/UCY training scenes plus filtered synthetic trajectories for up to 50 epochs at learning rate 10^{-4} , with early stopping and gradient clipping. The goal is to use SDD for broad motion statistics, ETH/UCY for target-domain realism, and synthetic trajectories for additional crowd configurations. This protocol also tests whether Vaibhav’s generated trajectories improve Muhammad’s pre-

TABLE II
MAIN PREDICTOR DESIGN CHOICES.

Model	Role in the comparison	Output
CV	Non-learned physical baseline and residual-model prior.	Mean plus noise
Social-LSTM	Recurrent social-pooling baseline from prior work.	Gaussian
SLSTM+V	Tests whether explicit velocity improves Social-LSTM.	Gaussian
Social-GRU	Tests bidirectional recurrent encoding under same social radius.	Gaussian
Transformer	Tests full-window attention over focal and neighbour tokens.	Gaussian
Diffusion	Tests stochastic sampling with a denoising trajectory model.	Samples plus Gaussian head
Trajectron++	Probabilistic graph-CVAE baseline for multi-modal coverage.	Samples/GMM

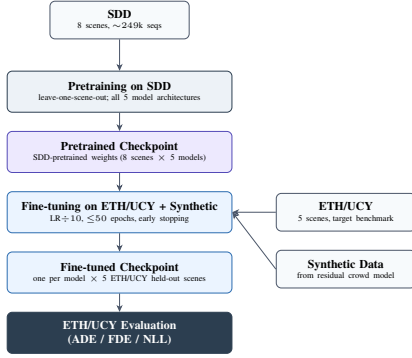


Fig. 5. Two-stage transfer: SDD pretraining followed by ETH/UCY and synthetic fine-tuning.

dictor training, instead of evaluating the generator only as a standalone component.

F. Downstream Safety Interface

The prediction stack is designed to feed a safety-aware navigation layer rather than operate as an isolated benchmark. In the intended integration, predictor residuals on a calibration split define conformal radii or regions, which can then be used by a planner or shield to reject unsafe actions. This follows the initial survey conclusion that prediction uncertainty must be preserved across module boundaries. Since the repository does not contain final closed-loop safety results, this report evaluates the prediction and world-model components and leaves full navigation validation as future work.

Benjamin’s component sits at this boundary. The nominal PPO/diffusion-policy planner is responsible for proposing robot actions, while the safety interface uses predicted pedestrian occupancies and calibrated residuals to decide whether the action is acceptable. The interface is modular: it can consume the Transformer’s Gaussian mean and residual calibration, Trajectron++ samples, or the diffusion predictor’s sampled futures. This lets the group test how prediction format affects downstream safety once full closed-loop experiments are run.

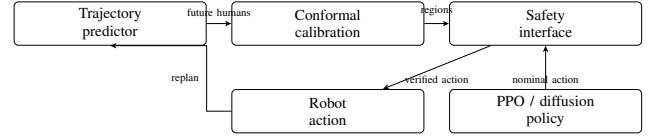


Fig. 6. Benjamin’s downstream safety-planning interface. The report evaluates the prediction inputs to this interface; closed-loop navigation metrics remain the next validation step.

TABLE III
DATASETS AND SPLITS USED IN THE REPORT.

Dataset	Use in this project	Split
ETH/UCY	Target pedestrian benchmark for all main predictor and residual-model results.	Leave-one-scene-out
SDD	Large-scale pretraining source and broader motion distribution for transfer learning.	Leave-one-scene-out
Synthetic	Social-CVAE generated trajectories filtered by turning behaviour and used for augmentation.	Train only

IV. EXPERIMENTAL SETUP

A. Synthetic Generation Setup

The Social-CVAE was pretrained on SDD and fine-tuned on ETH/UCY using 8 observed and 12 future timesteps. For each ETH/UCY seed trajectory, 20 futures were generated with latent sampling temperature 0.7. The quality-control test compared unfiltered generation with angle-based filtering. In the filtered setting, the model generated extra candidates with a multiplier of 5 and kept samples closest to the ETH/UCY turning-angle target of about 46° , within a 20° to 90° range.

Evaluation used both prediction and realism metrics. ADE, FDE, minADE@20, and minFDE@20 were computed in the normalised evaluation space. Speed, acceleration, and turning-angle statistics compared real and generated trajectories.

B. World Model Setup

The residual crowd world model was evaluated on a leave-one-scene-out ETH/UCY setup with eth held out. Training used the remaining ETH/UCY scenes and filtered synthetic trajectories. The comparison includes constant velocity, a Simple Residual Model using only the target trajectory, a Social Residual Model using neighbouring pedestrian context, and a Latent Crowd World Model with higher capacity. ADE and FDE were reported in metres.

C. Predictor Benchmark Setup

The main predictor benchmark uses ETH [11] scenes eth and hotel, UCY [12] scenes univ, zara1, and zara2, and SDD scenes from the Stanford Drone Dataset [2]. ETH/UCY is sampled at 2.5 Hz with 8 observed and 12 predicted steps. SDD is downsampled from 30 fps to 2.5 Hz using stride 12 and converted from pixels to metres at 0.0417 m/px.

The evaluation uses leave-one-scene-out splits. Metrics are ADE and FDE, which measure mean and final Euclidean error; minADE@20 and minFDE@20, which evaluate the best of 20 samples; and NLL for probabilistic Gaussian outputs. All

learned models use Adam with learning rate 10^{-3} , weight decay 10^{-4} , batch size 64, and ReduceLROnPlateau scheduling. Fine-tuning uses learning rate 10^{-4} , patience 15, and gradient clipping at 1.0.

The metrics answer different questions. ADE and FDE measure whether the most likely forecast is geometrically close to the true path. minADE@20 and minFDE@20 measure whether a model can sample at least one plausible future that matches the realised motion. NLL measures whether the model assigns probability mass appropriately, which matters for calibrated safety filtering. This multi-metric setup follows directly from the initial survey finding that no single forecasting score captures planning risk.

V. RESULTS AND DISCUSSION

A. Synthetic Data Generation Results

Table IV shows that the Social-CVAE can reconstruct and sample trajectories with very low normalised prediction error. The speed mean is close to the real ETH/UCY value, but acceleration variance is higher and generated turning angles are smoother than real trajectories.

TABLE IV
SYNTHETIC TRAJECTORY GENERATION RESULTS. LOWER ADE/FDE VALUES ARE BETTER.

Metric	Real	Synthetic / Model
ADE	—	4.61×10^{-4}
FDE	—	8.50×10^{-4}
minADE@20	—	4.59×10^{-4}
minFDE@20	—	7.72×10^{-4}
Speed Mean	7.49×10^{-4}	8.02×10^{-4}
Speed Std.	8.16×10^{-4}	1.13×10^{-3}
Acceleration Mean	6.33×10^{-4}	4.77×10^{-4}
Acceleration Std.	7.77×10^{-4}	1.74×10^{-3}
Turning Angle Mean	46.4°	28.8°
Turning Angle Std.	42.1°	40.3°

The main limitation is turning behaviour. The synthetic turning-angle mean is 28.8° , compared with 46.4° for real trajectories, which means the generator tends to produce smoother paths. Angle filtering improved the exported set’s mean turning angle from 15.9° to 21.4° in Vaibhav’s analysis, but it did not fully match the real distribution. The result supports a limited claim: filtered synthetic trajectories can expand training diversity, but real ETH/UCY trajectories must remain the anchor distribution.

The speed and acceleration statistics explain why the augmentation is still valuable. The generated speed mean, 8.02×10^{-4} , is close to the real mean, 7.49×10^{-4} , so the generator preserves the overall motion scale. At the same time, the acceleration standard deviation is higher than the real value, which indicates that the sampled set contains additional local motion variation. This variation needs quality control: too little would duplicate ETH/UCY, while too much would

Placeholder for Vaibhav’s generated-diversity plot

Fig. 7. Generated trajectory diversity from the same observed past. The placeholder is replaced when Vaibhav’s PNG is copied into the figures directory.

Placeholder for Vaibhav’s real-vs-synthetic distribution plot

Fig. 8. Real and synthetic motion distributions for speed, acceleration, turning angle, and nearest-neighbour distance.

corrupt the data distribution. The angle filter provides that control.

B. Residual Crowd World Model Results

The residual world model improves over constant velocity on held-out eth (Table V). The Simple Residual Model reduces ADE from 1.0755 m to 0.9514 m and FDE from 2.2819 m to 1.9171 m, giving relative improvements of 11.5 percent and 16.0 percent.

TABLE V
RESIDUAL CROWD WORLD MODEL RESULTS ON HELD-OUT ETH AFTER FINE-TUNING WITH REAL ETH/UCY AND FILTERED SYNTHETIC TRAJECTORIES.

Model	ADE (m)	FDE (m)	ADE Imp.	FDE Imp.
Constant Velocity	1.0755	2.2819	—	—
Simple Residual Model	0.9514	1.9171	11.5%	16.0%
Social Residual Model	1.0069	2.0752	6.4%	9.1%
Latent Crowd World Model	Worse than CV	Worse than CV	—	—

The result supports the residual formulation in Eq. (2). Constant velocity captures much of the local pedestrian motion, and the learned residual corrects the cases where people slow, turn, or deviate. The Social Residual Model also improves over constant velocity, but it underperforms the simpler model. This suggests that neighbour context was not automatically

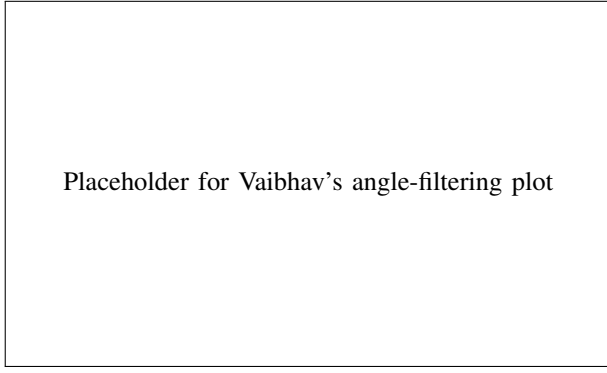


Fig. 9. Effect of angle-based filtering on synthetic trajectory statistics.

beneficial in the held-out eth split; the additional capacity may have learned scene-specific patterns from the training and synthetic data. The Latent Crowd World Model performing worse than constant velocity reinforces the same point: larger models do not automatically generalise better.

This negative result is important for system design. More social context does not guarantee better prediction; context quality and model capacity both matter. If neighbour identities are sparse, noisy, or only weakly informative, the social branch can add variance without adding predictive signal. For a downstream planner, the simple residual model has a clearer failure mode: it stays close to constant velocity unless the target history supports a correction. This behaviour is easier to combine with safety filtering than a larger model that may produce confident but poorly generalised deviations.

The limitation is that the evaluation uses one held-out scene. The result should not be treated as a general claim of world-model superiority until the comparison is repeated across all five ETH/UCY splits and includes a real-only versus real-plus-synthetic ablation.

The training curves in Fig. 10 are included for this reason. Both residual variants reached their best validation behaviour early, after which training error could continue improving without reliably improving held-out performance. Early stopping is therefore part of the generalisation strategy for a small-scene pedestrian benchmark.

C. ETH/UCY Predictor Results

For context, published state-of-the-art results on ETH/UCY include PECNet [13], AgentFormer [14], and LED [15], which report minADE@20 in the 0.18–0.27 m range under heavier compute budgets and goal-conditioned designs. The benchmark here is not aimed at beating those numbers; it asks which architectures balance point accuracy, diversity, and calibration in a way that supports downstream conformal safety filtering.

Table VI compares all trajectory predictors on ETH/UCY. The Transformer gives the lowest ADE at 0.568 m and the lowest NLL at 1.54. The diffusion model gives the lowest FDE at 1.195 m. Trajectron++ gives the best minADE@20 at 0.385 m, showing that its CVAE latent variable covers diverse future modes better than the other models.

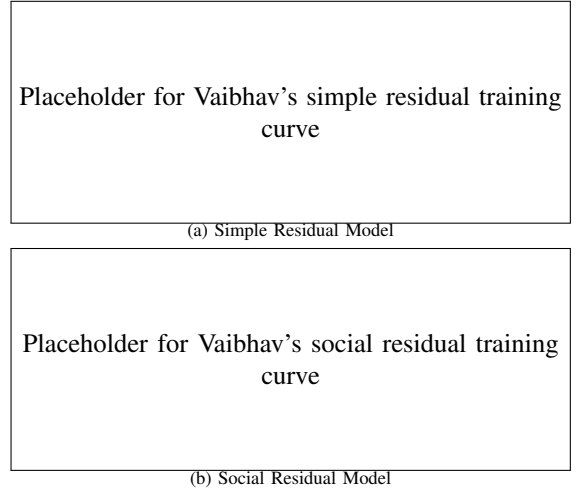


Fig. 10. Training behaviour of residual world models on the held-out eth split. The placeholders are replaced when Vaibhav's PNGs are copied into the figures directory.

TABLE VI
ETH/UCY LEAVE-ONE-OUT RESULTS AVERAGED OVER FIVE SCENES.
MADE/MFDE DENOTE MINADE@20/MINFDE@20. BEST VALUES ARE BOLD.

Model	ADE	FDE	mADE	mFDE	NLL
CV Baseline	0.736	1.272	0.610	0.814	—
Social-LSTM [3]	0.596	1.248	0.581	0.597	18.0
SLSTM+V	0.585	1.240	0.587	0.585	36.2
Social-GRU	0.586	1.238	0.596	0.924	—
Transformer	0.568	1.203	0.638	0.532	1.54
Diffusion	0.579	1.195	1.174	2.032	5.14
Trajectron++ [10]	0.853	1.817	0.385	0.674	8.49

These results show why one metric is not enough. ADE rewards the most accurate central prediction, where the Transformer performs best. minADE@20 rewards having at least one sample close to the true future, where Trajectron++ is best. For safety-aware navigation, both matter: a small average error can shrink conformal radii, but missing plausible alternative futures can still create unsafe overconfidence.

The NLL result is also important. The Transformer's NLL of 1.54 is much lower than Social-LSTM's 18.0 and Trajectron++'s 8.49 in our runs. This does not mean the Transformer is globally better for navigation, because it still has weaker best-of-20 diversity than Trajectron++. It does mean that, under the Gaussian head used here, the Transformer gives a sharper and better calibrated single-distribution forecast. A downstream conformal layer would still calibrate on residuals, but starting from a smaller and better shaped residual distribution can reduce conservatism.

Velocity augmentation gives a small but consistent gain over Social-LSTM: average ADE falls from 0.596 m to 0.585 m. Social-GRU nearly matches Social-LSTM, suggesting that the shared social-pooling radius may be more limiting than the recurrent cell choice. The diffusion predictor becomes competitive after sharing the encoder with the Gaussian head, but its best-of-20 diversity remains weak, so its samples cluster

TABLE VII
PER-SCENE ETH/UCY ADE. BEST VALUE PER SCENE IS BOLD.

Scene	CV	SLSTM	SLSTM+V	T++	Trans.	Diff.
eth	1.204	1.015	1.010	1.355	0.982	0.982
hotel	0.552	0.541	0.527	0.977	0.437	0.485
univ	0.716	0.655	0.635	0.772	0.568	0.563
zara1	0.623	0.447	0.432	0.624	0.481	0.503
zara2	0.583	0.319	0.320	0.535	0.371	0.361

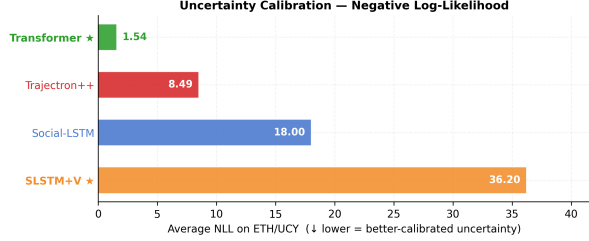


Fig. 11. Average NLL comparison on ETH/UCY for probabilistic predictors. Lower values indicate better likelihood under the reported output distribution.

around the mean rather than covering alternative futures.

Scene-level behaviour matters. The eth scene is hardest for almost every model, with ADE near or above 1 m, while zara2 is much easier because motion is more regular. This affects system deployment because calibration sets should match the target environment. A safety radius calibrated mostly on easy scenes would likely be too small for eth-like movement; a radius calibrated only on the hardest scene would be safe but more conservative elsewhere.

The per-scene table also prevents overclaiming from the average alone. The Transformer has the best average ADE, but it is not best on every scene: SLSTM+V performs best on zara1, Social-LSTM narrowly wins on zara2, and diffusion matches or beats the Transformer on eth and univ. This variation matters for the final safety system because it suggests no predictor should be trusted without scene-aware calibration.

The diffusion ablation is one of the clearest implementation lessons. The initial detached-encoder design separated the Gaussian head from the denoising training signal too strongly, producing poor point prediction. Sharing the encoder restored ADE and FDE, but minADE@20 stayed high: the fix made diffusion accurate around the dominant future without producing the broader sample coverage that Trajectron++ does.

D. SDD Results

Table IX reports SDD leave-one-out results. Every learned model improves over constant velocity by a wide margin. SLSTM+V gives the best ADE, minADE@20, and minFDE@20, while the Transformer gives the best FDE. The diffusion model again has poor diversity, with minADE@20 equal to 1.710 m.

SDD is larger and more varied than ETH/UCY, so the results also test whether the models remain stable at scale. The recurrent models benefit more from explicit velocity on SDD than on ETH/UCY, likely because SDD includes more turning and speed variation. Transformer performance remains

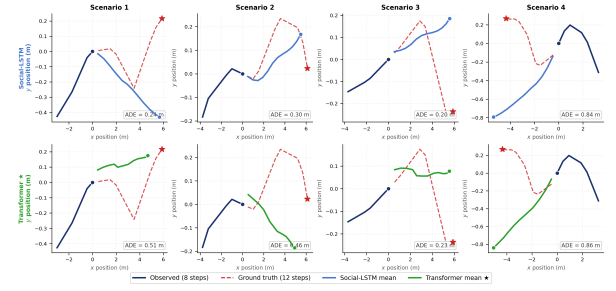


Fig. 12. Qualitative prediction on zara1. The examples show that per-scene behaviour can differ from the aggregate ETH/UCY ranking.

TABLE VIII
DIFFUSION ABLATION ON ETH/UCY. REMOVING THE ENCODER DETACH MADE THE DIFFUSION MODEL USABLE AS A PREDICTOR WITHOUT IMPROVING DIVERSITY.

Version	ADE	FDE	minADE@20	NLL
Detached encoder, $\lambda = 1.0$	1.370	2.539	1.143	3.31
Shared encoder, $\lambda = 0.3$	0.579	1.195	1.174	5.14

strong but no longer separates from the recurrent models as much as it does on ETH/UCY.

The SDD results also help explain why transfer is not uniformly beneficial. SDD supplies many more trajectories, but its aerial viewpoint, broader scene types, and mixed agent behaviours do not perfectly match ETH/UCY. Pretraining therefore learns transferable low-level motion patterns while still requiring target-domain fine-tuning before evaluation on ETH/UCY.

E. Transfer and Fine-Tuning

Table X compares scratch ETH/UCY training with SDD pretraining followed by ETH/UCY and synthetic fine-tuning. The recurrent models improve by 5 to 10 percent. The Transformer barely changes, moving from 0.568 m to 0.570 m. Diffusion improves on ADE but not on diversity.

This supports a practical conclusion: transfer helps most when the architecture still needs broad motion regularisation. The recurrent models use SDD as a regulariser and then adapt to ETH/UCY. The Transformer already extracts most of the relevant pattern from the small ETH/UCY training set, so the extra data does not help. This aligns with transfer-learning evidence that more source data helps only when the source and target structure match the model being fine-tuned [16].

F. Integrated Discussion

The three completed parts fit together. Synthetic data broadens the training support, but must be filtered because unfiltered generation can distort turning statistics. The residual world model gives a simple planning-facing motion model, and the held-out eth result shows that a learned correction over constant velocity improves prediction. Muhammad’s predictor benchmark then separates point accuracy, diversity, and calibration: the Transformer is the best point predictor and

TABLE IX
SDD LEAVE-ONE-OUT RESULTS AVERAGED OVER EIGHT SCENES. BEST VALUES ARE BOLD.

Model	ADE	FDE	mADE	mFDE	NLL
CV Baseline	0.926	1.620	0.797	1.154	—
Social-LSTM	0.672	1.358	0.783	0.621	1.162
SLSTM+V	0.656	1.340	0.755	0.604	1.289
Social-GRU	0.667	1.379	0.793	0.625	1.331
Transformer	0.657	1.326	0.808	0.633	1.474
Diffusion	0.673	1.335	1.710	3.159	1.683

TABLE X
SDD PRETRAINING THEN ETH/UCY AND SYNTHETIC FINE-TUNING.
VALUES ARE ETH/UCY ADE IN METRES AVERAGED OVER FIVE SCENES.

Model	Scratch	SDD then Synth FT	Δ (%)
Social-LSTM	0.596	0.537	−9.9
SLSTM+V	0.585	0.555	−5.1
Social-GRU	0.586	0.549	−6.3
Transformer	0.568	0.570	+0.4
Diffusion	0.579	0.535	−7.6

gives the lowest NLL, while Trajectron++ is better at covering multiple possible futures.

The open problem is closed-loop validation. A conformal or PPO safety-planning layer cannot simply consume the best ADE model and assume safety follows. Prediction errors must be calibrated, uncertainty must be composed over agents and timesteps, and the resulting safety sets must still leave feasible motion. This is exactly why the world model and predictor benchmark should be treated as components of a navigation stack, not as isolated leaderboards.

G. How The Components Answer The Objectives

The original project objective was safe and efficient navigation in human crowds. The completed work does not yet prove closed-loop navigation safety, but it builds the part of the stack that determines what the planner believes about people. That belief is central: if the predicted pedestrian motion is wrong, every downstream safety decision is made from the wrong state of the world. The current report therefore answers a narrower but necessary question: which prediction and world-model components are suitable inputs to a safety aware navigation system?

The synthetic generator addresses data scarcity. ETH/UCY is valuable because it is a standard benchmark, but it is too small to cover the crowd configurations a robot might encounter. The Social-CVAE gives a way to generate additional future trajectories conditioned on observed motion. Its limitation, smoother turning behaviour, is not a failure of the idea; it is evidence that synthetic data needs distributional checks before being trusted, which is why the report evaluates speed, acceleration, and turning statistics rather than only ADE.

The residual world model addresses planning speed and interpretability. A planner does not always need a heavy distributional model at every control cycle. It may need a

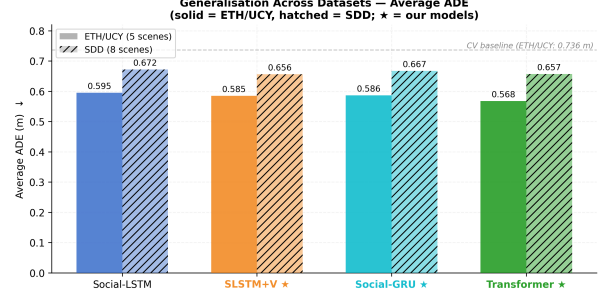


Fig. 13. Fine-tuning comparison showing that SDD pretraining helps recurrent models more than the Transformer under the current protocol.

fast estimate of how nearby pedestrians will move if the robot proposes an action. The residual formulation is attractive because its failure mode is easier to reason about than a fully learned black-box forecast: the output equals constant velocity plus a correction. When the correction becomes large, the planner or safety layer can treat the forecast as less trustworthy.

The predictor benchmark addresses accuracy, diversity, and calibration. The Transformer is a strong candidate when the downstream system wants a sharp mean forecast and compact uncertainty. Trajectron++ is a stronger candidate when the downstream system needs to cover multiple plausible futures. The diffusion model provides an implementation lesson: stochastic generation alone does not guarantee meaningful diversity. If conditioning overwhelms the denoising process, the model may generate many samples clustered around the same mean.

Benjamin’s safety/planning interface ties these findings back to robot control. For the interface, the key question is not simply which model has the lowest ADE. A PPO policy may propose a smooth and efficient action, but the safety interface needs to know whether that action remains safe under the forecasted pedestrian uncertainty. The predictor format changes this check. A Gaussian predictor supports residual-radius calibration; a sample-based model supports hulls or empirical occupancy; a multi-modal GMM supports level-set style reasoning. The report therefore prepares the comparison Benjamin needs for choosing which predictor output to connect to the planner.

H. Planned Closed-Loop Evaluation

The next experimental step is to run the same predictor choices inside a navigation simulator. Each episode should place the robot in a start-goal task with moving pedestrians. At each control cycle, the predictor receives the recent pedestrian histories, the nominal PPO/diffusion policy proposes an action, and the safety interface either accepts that action or replaces it with a safer one. This setup would directly test whether the open-loop differences reported here matter for robot behaviour.

The first closed-loop metric should be collision rate. It is the most basic safety measure, but it is not sufficient by itself. A robot that never collides because it refuses to move has failed the navigation task. The second metric should therefore

be success rate or time-to-goal. The third should be freezing rate, measured as the fraction of time the robot remains nearly stationary despite having a reachable goal. The initial survey identified conservatism as a common failure mode of safer systems.

The fourth metric should be intervention frequency. If the safety interface overrides the nominal policy at nearly every timestep, the predictor and policy are not aligned. If it never intervenes, the safety layer may be too weak or the test scenarios too easy. A successful system should have low collisions, reasonable time-to-goal, and interventions concentrated in the scenes that actually pose risk.

The fifth metric should be achieved control rate. A predictor that is accurate but slow can cause stale decisions. The benchmark should therefore report the end-to-end time per control cycle, not only the model forward-pass time. This will make it possible to compare the lightweight residual world model, Transformer predictor, Trajectron++ samples, and diffusion samples under the same real-time budget.

Finally, the closed-loop evaluation should include degraded sensing. Occlusions and missed detections are normal in crowds. Testing only with perfect pedestrian tracks would overstate safety. A practical evaluation should remove or perturb some pedestrian observations and then measure whether the conformal safety interface remains robust or becomes overly conservative.

I. Limitations and Remaining Work

Several limitations affect the strength of the conclusions. First, ETH/UCY is a small benchmark. Leave-one-scene-out testing helps, but five scenes cannot cover the full variety of real crowd behaviour. The synthetic generator addresses data scarcity, yet its smoother turning distribution shows that synthetic data can also introduce bias. A stronger augmentation study would compare real-only, real-plus-unfiltered-synthetic, and real-plus-filtered-synthetic training across all ETH/UCY scenes.

Second, the residual world model was evaluated only on held-out eth in the current report. The result is promising because it improves over constant velocity, but it should be repeated across hotel, univ, zara1, and zara2 before being treated as a general dynamics model. The fact that the social residual model underperforms the simpler residual model also deserves more analysis: neighbour context may help only in specific density regimes, or the pooling method may be too crude to represent interaction cues.

Third, the predictor benchmark is open-loop. It measures future trajectory accuracy given observed pedestrian histories, but it does not yet measure what happens when a robot acts on those forecasts. In closed loop, a small prediction delay, a missed pedestrian, or a conservative conformal radius can change navigation behaviour. The next evaluation should therefore report collision rate, time-to-goal, intervention frequency, planner latency, and freezing rate with Benjamin’s safety interface connected.

Finally, the current work treats perception as already solved. Real robots must track people under occlusions, noisy detections, identity switches, and limited field of view. Those errors would change both the predictor input and the safety calibration residuals. A deployment-ready system should therefore calibrate not only trajectory prediction error, but also tracking and execution error.

VI. CONCLUSIONS

This report presents the design and evaluation work completed so far for safe robot navigation in human crowds. Vaibhav’s Social-CVAE generator produces synthetic trajectories that can augment ETH/UCY, but the generated turning statistics show why filtering is necessary. His residual crowd world model improves over constant velocity on held-out eth, reducing ADE by 11.5 percent and FDE by 16.0 percent. This supports its use as a lightweight dynamics model for planning-facing pedestrian forecasts, although the evidence should be extended to all ETH/UCY splits.

Muhammad’s predictor benchmark shows that model choice depends strongly on the metric. The Transformer gives the best ETH/UCY point accuracy and NLL, with ADE 0.568 m and NLL 1.54. Trajectron++ gives the best multi-modal coverage, with minADE@20 0.385 m. Diffusion becomes competitive on ADE/FDE after the shared encoder fix, but its samples still lack diversity. SDD pretraining improves recurrent models by 5 to 10 percent, while the Transformer gains little.

The main lesson is that prediction quality is multi-dimensional. A safe crowd navigation system needs data diversity, a planning-facing world model, accurate point forecasts, calibrated uncertainty, and closed-loop validation. The next step is to connect these prediction outputs to Benjamin’s conformal/PPO safety-planning interface and evaluate the full stack under collision, efficiency, latency, and conservatism metrics.

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